EXPERIMENT 15.1.1

ALGORITHM: Car Dealership System using OOP

Step 1: Start

Step 2: Define a base class Car

Attributes:

brand

price

model

color

Step 3: Define method display_details() in Car

Print brand

Print price

Print model

Print color

Step 4: Define derived class Car1 inheriting from Car

Override display_details() method

Print brand

Print price

Print model

Print color

Step 5: Define derived class Car2 inheriting from Car

Override display_details() method

Print brand

Print price

Print model

Print color

Step 6: In main program

Read first line input for Car1

Split into:

brand1, price1, model1, color1

Read second line input for Car2

Split into:

brand2, price2, model2, color2

Step 7: Create object c1 of class Car1

Assign values:

c1.brand = brand1

c1.price = float(price1)

c1.model = model1

c1.color = color1

Step 8: Create object c2 of class Car2

Assign values:

c2.brand = brand2

c2.price = float(price2)

c2.model = model2

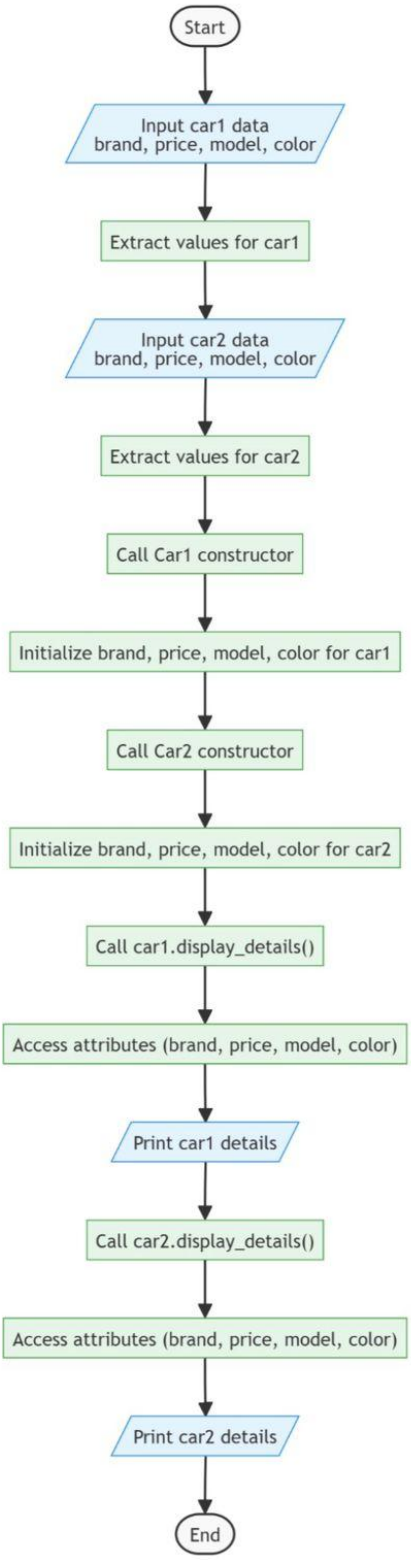
c2.color = color2

Step 9: Call c1.display_details()

Step 10: Call c2.display_details()

Step 11: Stop

FLOWCHART:


Start
Input car1 data
brand, price, model, color
Extract values for car1
Input car2 data
brand, price, model, color
Extract values for car2
Call Car1 constructor
Initialize brand, price, model, color for car1
Call Car2 constructor
Initialize brand, price, model, color for car2
Call car1.display_details()
Access attributes (brand, price, model, color)
Print car1 details
Call car2.display_details()
Access attributes (brand, price, model, color)
Print car2 details
End

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15.1.1. Car Dealership Sales System 11:10

Write a Python program to model a car dealership sales system using object-oriented programming. Create a base class **Car** and two derived classes **Car1** and **Car2** to represent different car types.

Class Structure

1. Base Class **Car**:
   - Attributes: **brand**, **price**, **model**, **color.**
   - Method: **display_details()** to print all car attributes.
2. Derived Classes **Car1** and **Car2**:
   - Inherit from **Car** class.
   - Override **display_details()** method to display car information in the specified format.

Input Format:

- The first line contains **Car1**'s **brand** (string), **price** (float), **model** (string), and **color** (string) separated by spaces.
- The second line contains **Car2**'s **brand** (string), **price** (float), **model** (string), and **color** (string) separated by spaces.

Explorer carDetail... Submit Debugger

```python
class Car:
    def __init__(self, brand, price, model, color):
        self.brand = brand
        self.price = price
        self.model = model
        self.color = color

def display_details(self):
    pass
class Car1(Car):
    def display_details(self):
        print(self.brand)
        print(self.price)
        print(self.model)
        print(self.color)
class Car2(Car):
    def display_details(self):
        print(self.brand)
        print(self.price)
```

Terminal Test cases